THE CYLINDER

SOLID MENSURATION

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%	30%	**40%**
PRELIM	MIDTERM	ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

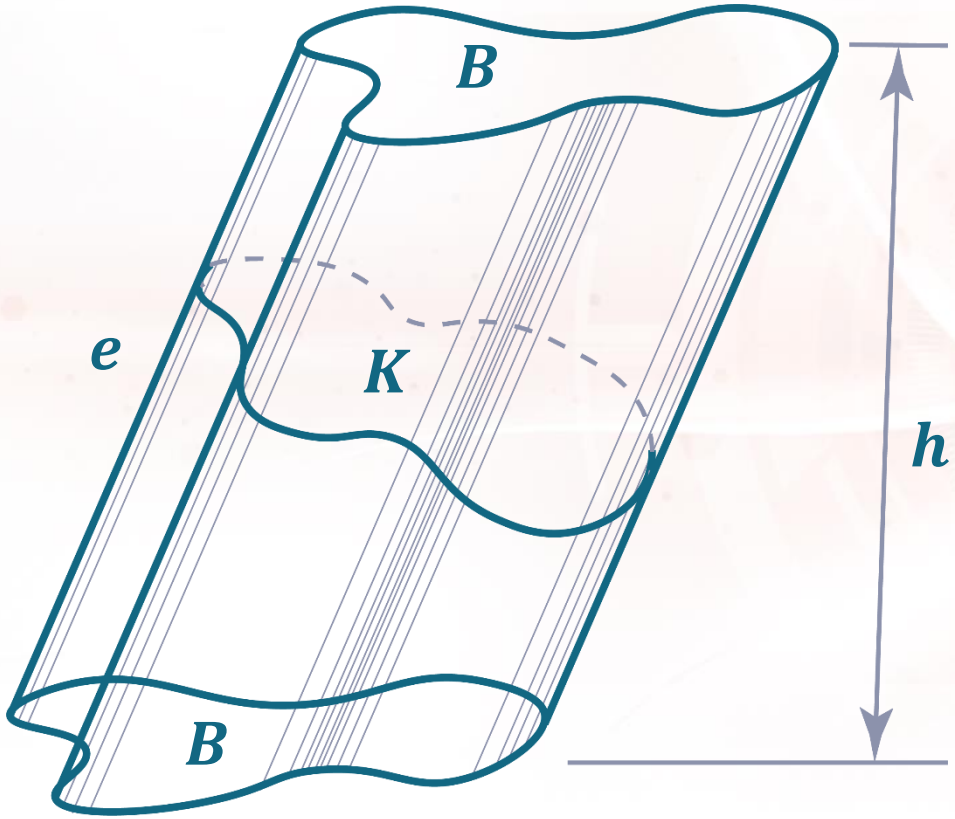
TOPIC OUTLINE

Properties of a Cylinder

- Lateral Surface Area
- Total Surface Area
- Volume

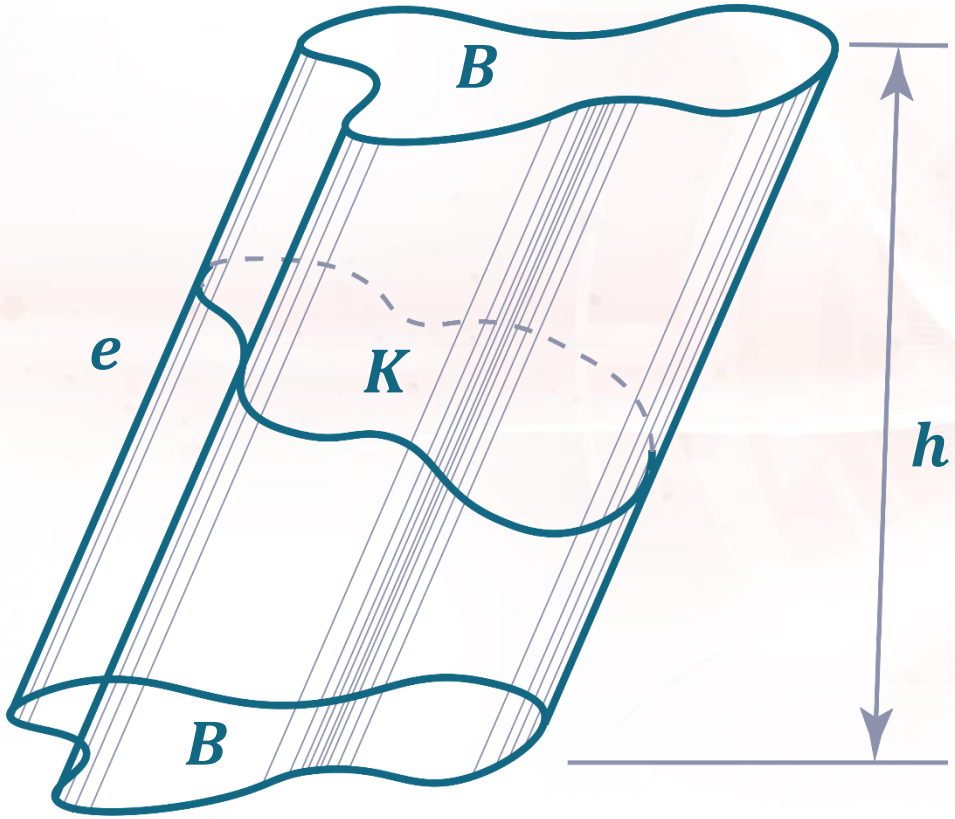
PROPERTIES OF A CYLINDER

DEFINITION



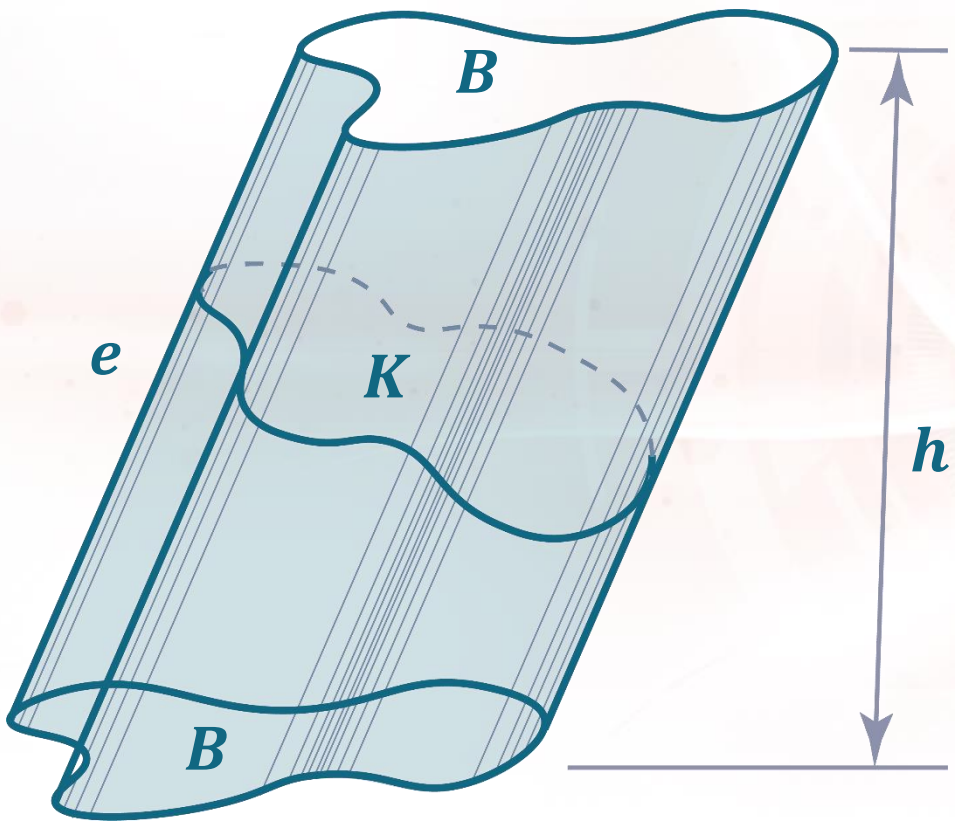
A **cylinder** is a solid bounded by a closed **cylindrical surface** and two parallel planes

PROPERTIES



1. The **bases, *B*** of a cylinder are equal.
2. The **altitude, *h*** of a cylinder is the perpendicular distance between the bases.
3. The **right section, *K*** of a cylinder is a section perpendicular to all elements of the cylinder.
4. The **elements, *e*** of a cylinder are equal.

AREA OF A CYLINDER



The Lateral Surface Area, **S**

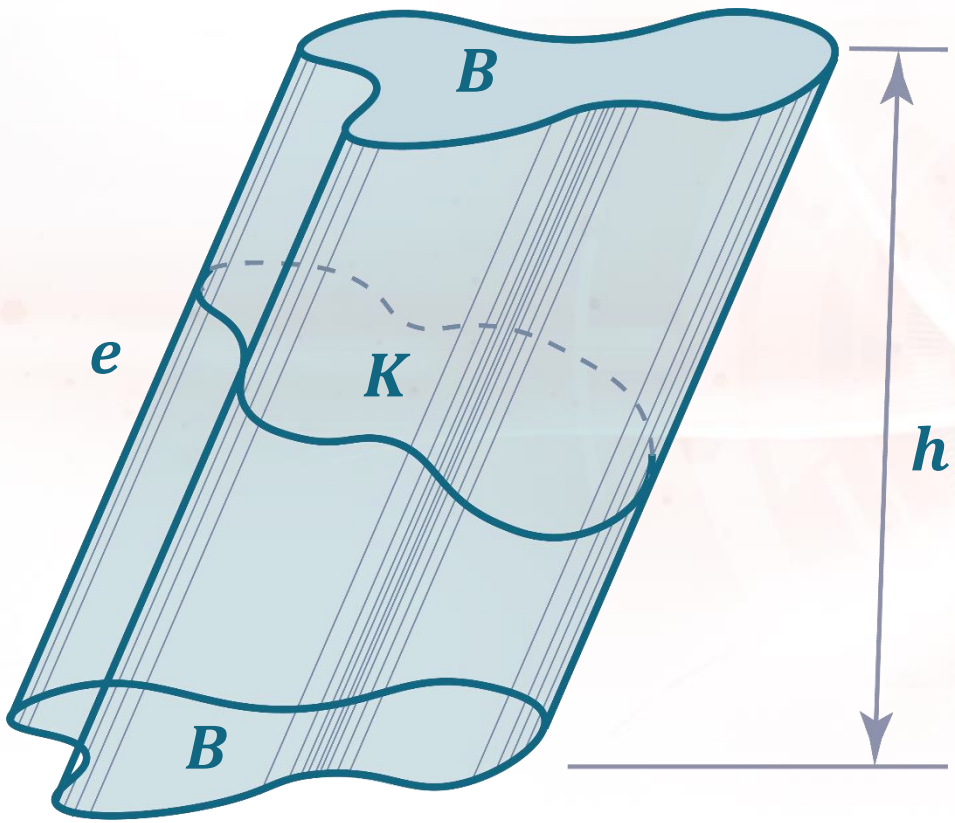
$$S = eP$$

where,

e = lateral edge

P = perimeter of right section

TOTAL SURFACE AREA



The Total Surface Area, **T**

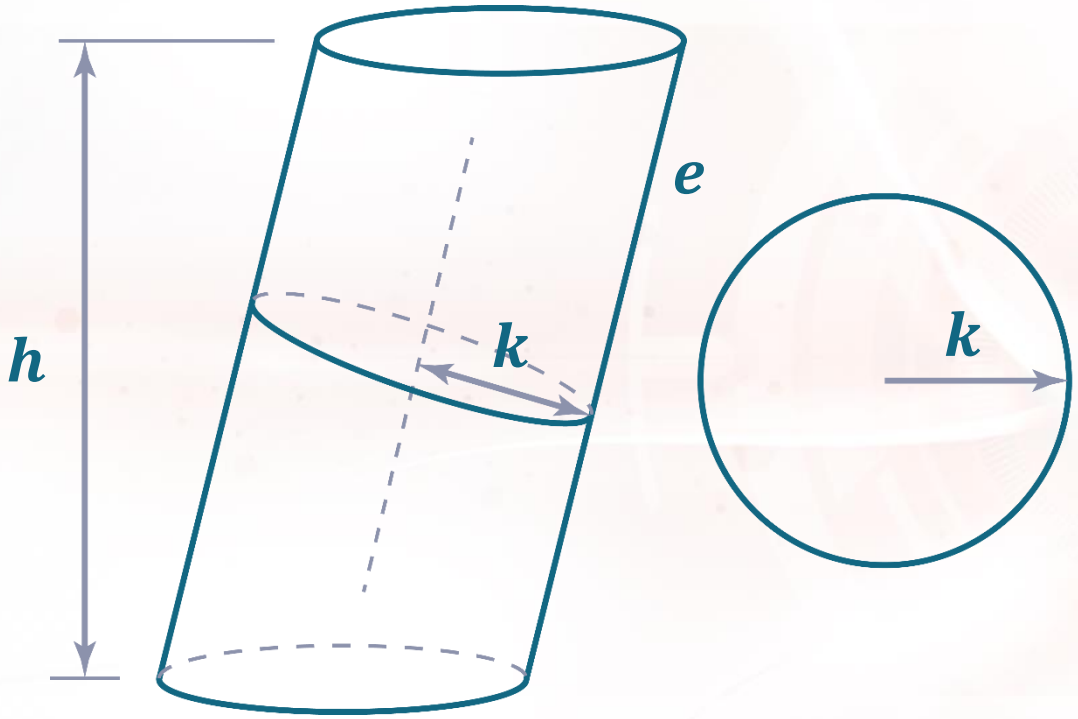
$$T = 2B + S$$

where,

B = area of either base

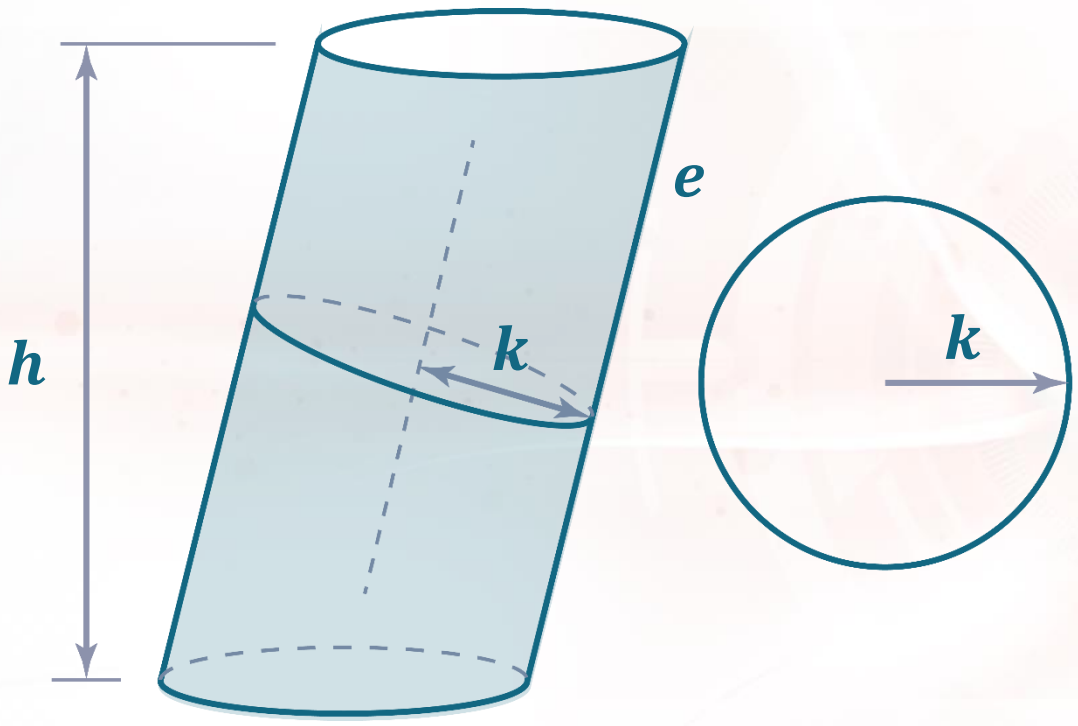
S = lateral surface area

CIRCULAR CYLINDER



A circular cylinder is a cylinder which has a **circular right section**

LATERAL SURFACE AREA



The Lateral Surface Area, **S**

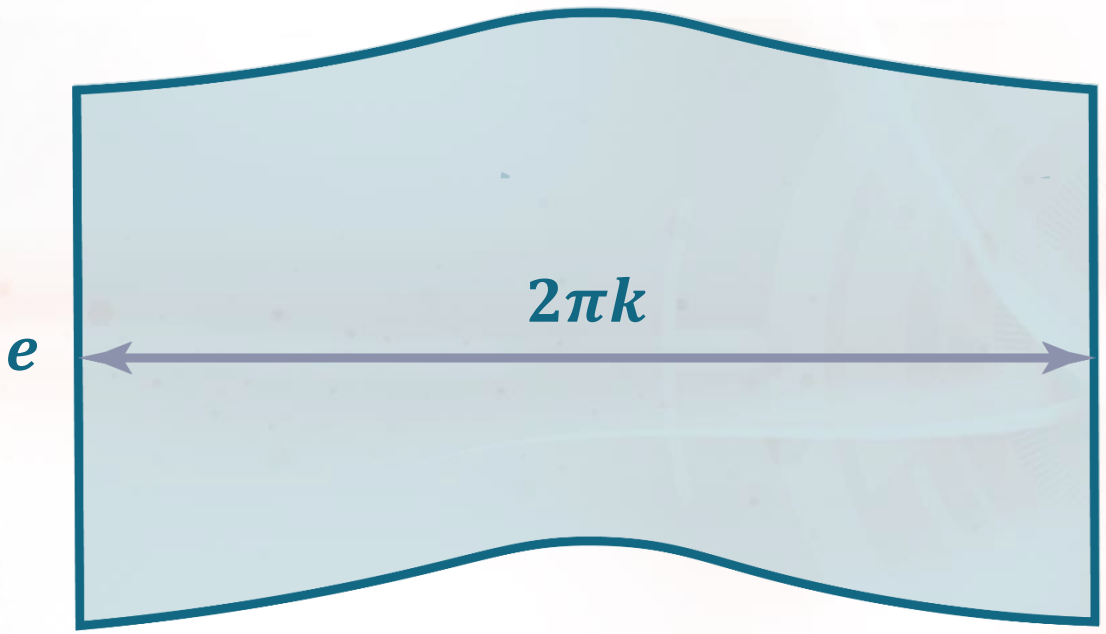
$$S = eP$$

$$S = 2\pi ke$$

where,

k = radius of right section

$e = element$



The Lateral Surface Area, **S**

$$s = eP$$

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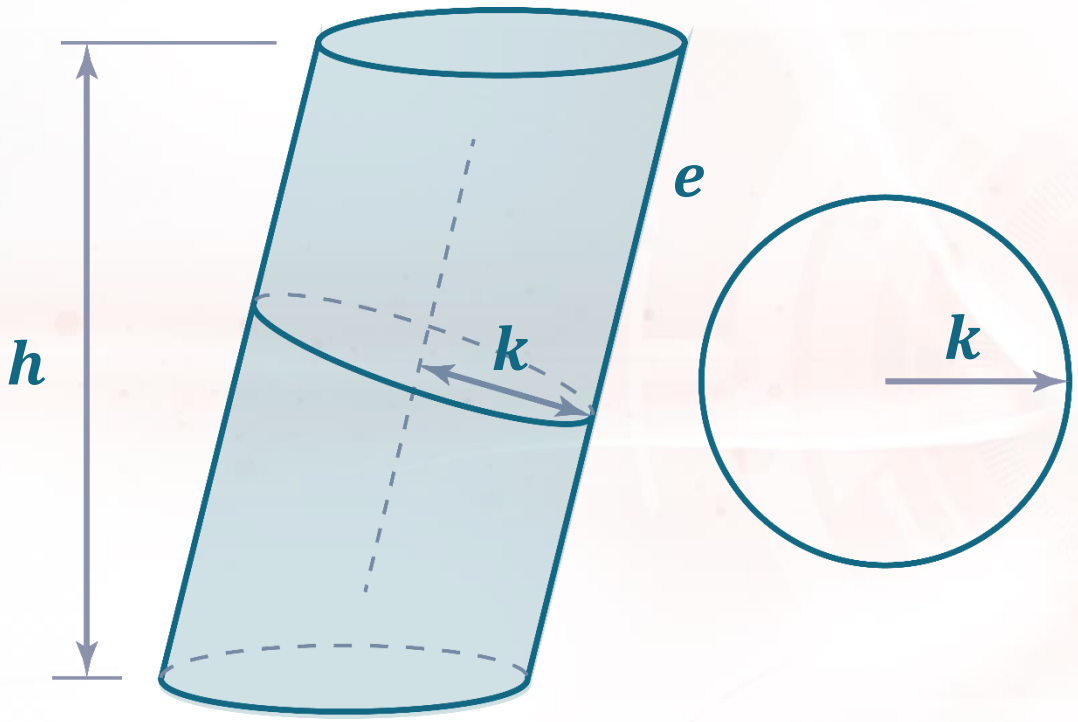


The Total Surface Area, **T**

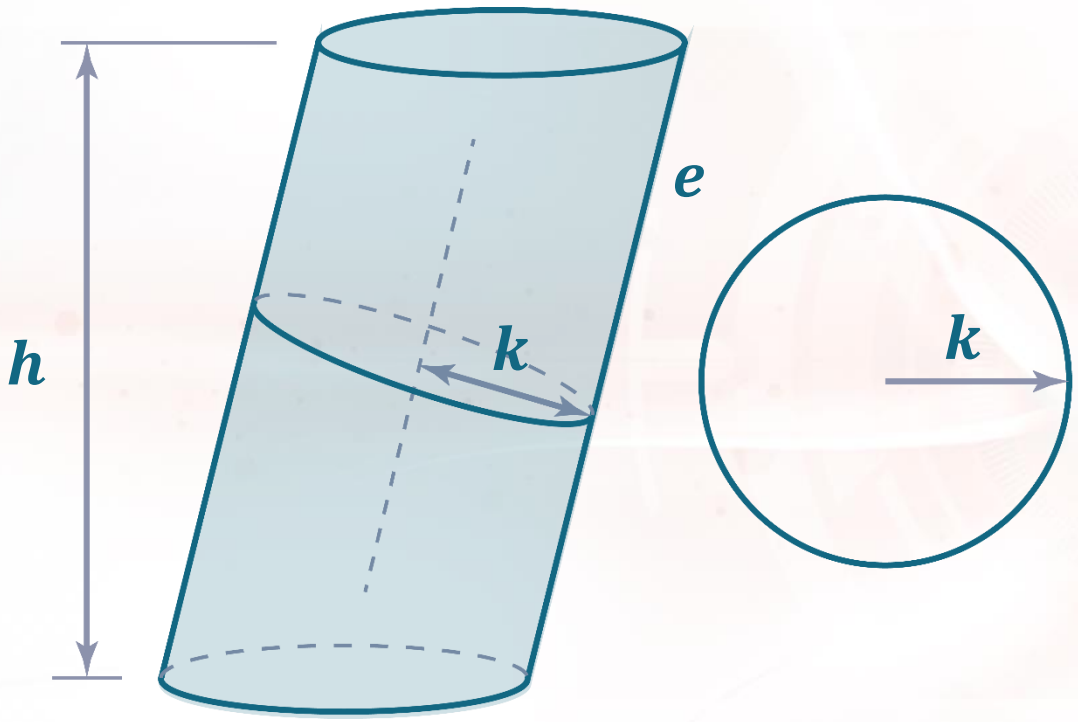
$$T = 2B + S$$

where,

B = area of either base

S = lateral surface area

Consider a metal **pipe** with a **radius** of **4** inches and a **height** of **10** inches. Calculate the **cost** of metal required to make it, assuming that the metal costs **$0.05 per square inch**.



The Volume, **V**

$$V = Bh$$

$$V = \pi k^2 e$$

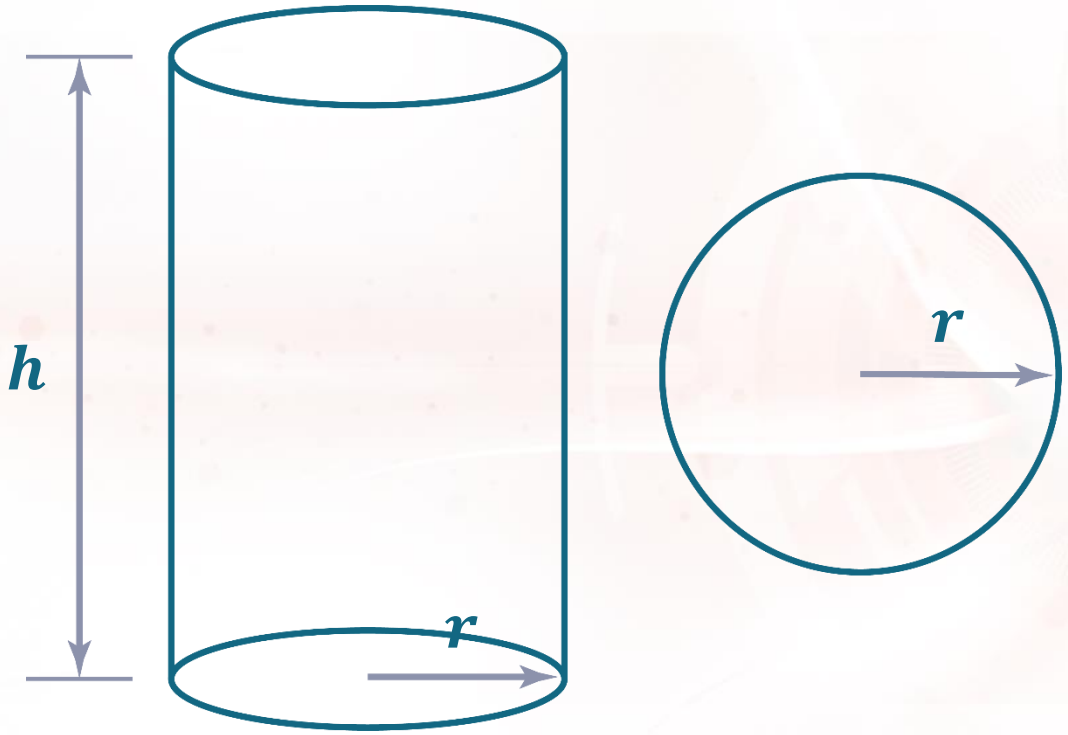
where,

$k = radius\ of right\ section$

$e = element$

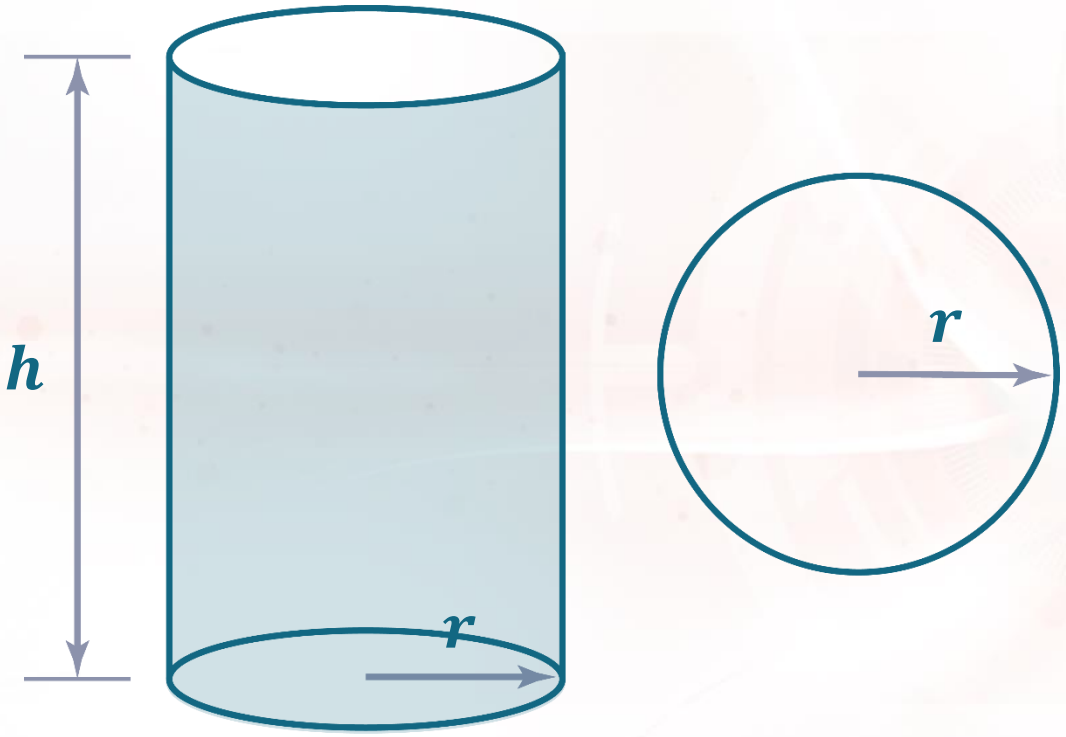
Consider a **cylindrical** container that is **12** inches **tall** and has a **radius** of **6** inches. Calculate the **volume** of the container, as well as the **cost** of filling it with water, assuming that the cost of water is **$0.01 per cubic inch**.

RIGHT CIRCULAR CYLINDER



A **right** circular cylinder is a cylinder whose elements are **perpendicular** to its base and whose bases are circles

LATERAL SURFACE AREA



The Lateral Surface Area, **S**

$$S = eP$$

$$S = 2\pi rh$$

where,

r = radius of base

$h = altitude$

The Lateral Surface Area, **S**

$$S = eP$$

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r = radius of base

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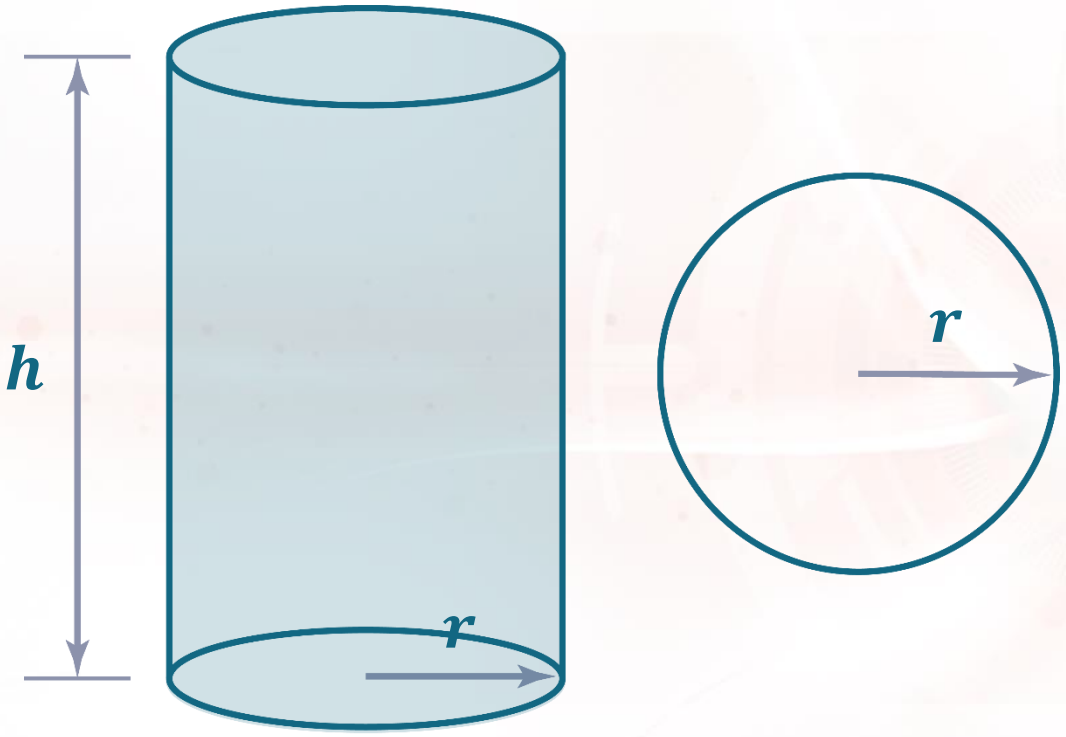
TOTAL SURFACE AREA



The Total Surface Area, **T**

$$T = 2B + S$$

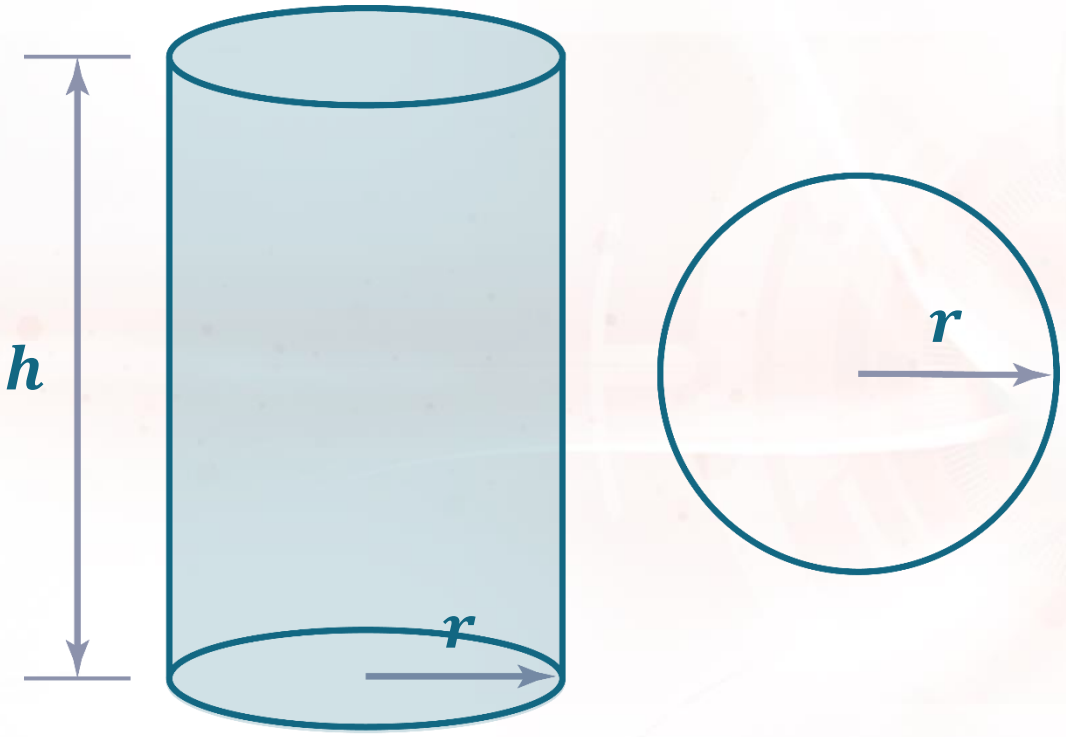
$$T = 2\pi r(h + r)$$

where,

r = radius of base

h = altitude

Consider a **cylindrical** pipe with a **diameter** of **8** inches and a **height** of **24** inches. The pipe has a cylindrical **hole** drilled straight through the center with a **diameter** of **4** inches that runs the entire height of the pipe. Calculate the **cost** of painting the pipe assuming the cost of paint is **$0.10 per sq. in**.



The Volume, **V**

$$V = Bh$$

$$V = \pi r^2 h$$

where,

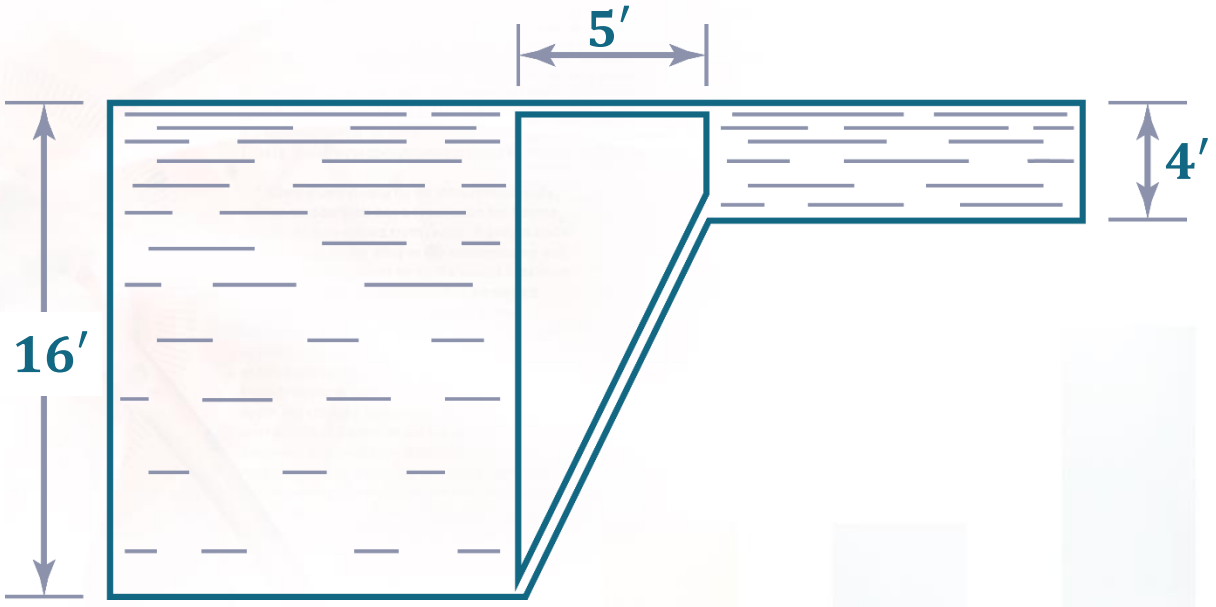
k = radius of base

h = altitude

Consider a hollow **right circular cylinder** with an **outer** radius of **10** cm, an **inner** radius of **8** cm, and a **height** of **20** cm. Calculate the **cost** of metal used to make it, assuming that the cost of the metal is **$0.05 per cubic centimeter**.

SEATWORK

1. Find the **volume** of the largest **cylinder** with **circular** base that can be inscribed in a **cube** whose **volume** is **27 cu.in**.

2. A pine log **40 ft. long** has a uniform circular cross section of **radius 11 in.** The ends lie in parallel planes which are **not** perpendicular to the axis of the log. How much **paint** will be required to give the log (ends not included) two coats, if **1 gal.** of paint covers **800 sq. ft.** of surface?

3. Two vertical brine tanks, with tops on the same level, one **16** ft. **deep**, the other **4** ft. **deep**, have their tops and bottoms connected by **pipes 3 in**. in **diameter**. If the pipe connecting the **tops** measures **5** ft., find the **weight** of brine in the **other pipe** when entirely full. (The brine weighs **66.8 lb. per cu. ft.**).